

EMANUEL PAULO

Backend Software Engineer | Java / Spring Boot / Cloud-Native / Microservices
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PROFESSIONAL SUMMARY

Backend engineer with 2 years of professional experience building cloud-native microservices in the automotive and banking sectors. Deep hands-on knowledge of Java, Spring Boot, Docker, and RESTful API design. Proven ability to own features end-to-end from technical design and implementation to testing, deployment, and production operation. Experienced in agile, cross-functional teams working in English. Actively learning Kotlin as a natural extension of my JVM background.

EXPERIENCE

Backend Software Engineer | Asseco PST, Banking Domain (Lisbon)

May 2025 – Present

- Contributed to the migration of a legacy core banking system to a modern Spring Boot microservices architecture, redesigning service boundaries and ensuring backward compatibility during a live production transition.
- Implemented critical financial features including transactional workflows, balance visibility controls, and maker-checker approval flows, ensuring data integrity, auditability, and compliance with European banking regulations.
- Designed and optimized SQL schemas and stored procedures for high-volume banking operations across SQL Server, achieving ~30% improvement in query performance under peak load.
- Built secure RESTful APIs with role-based authentication and authorization, structured logging, and full OpenAPI documentation, supporting integrations with internal and external financial systems across multiple European markets.
- Wrote unit and integration test suites using JUnit 5 and Testcontainers, significantly improving test coverage and enabling confident refactoring of core transaction logic.
- Automated CI/CD pipelines with GitLab, reducing deployment time by ~25%, enabling daily production deployments and improving release reliability.
- Containerized backend services with Docker and contributed to environment configuration, ensuring parity between development, staging, and production.
- Participated in technical design discussions, code reviews, and sprint planning within an agile team, contributing to architecture decisions and API design standards.

Backend Software Engineer | Critical Techworks, BMW Group (Lisbon)

April 2024 – April 2025

- Designed and developed high-performance microservices using Java and Quarkus for BMW's digital platform, handling real-time automotive telemetry and ownership data across 1,000+ daily transactions.
- Implemented reactive and asynchronous processing with the Mutiny API, improving throughput and responsiveness for real-time data pipelines — directly applicable to Spring Reactive Stack patterns.
- Built and maintained RESTful APIs integrated with multiple BMW internal systems, ensuring robust error handling, versioning, and clear consumer contracts.
- Optimized data access using Panache ORM and custom SQL queries, improving system performance and data consistency across distributed services.
- Established CI/CD pipelines with Docker that automated testing and reduced deployment time by 30%, enabling faster, safer releases.
- Delivered secure, well-documented services in an international, cross-functional team working fully in English across engineering, product, and architecture disciplines.

EDUCATION

Bachelor in Software Engineering | Universidade Aberta, Lisbon | Dec 2025 – Jul 2028

TECHNICAL SKILLS

Languages: Java, Kotlin (learning), SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring MVC, Quarkus, Hibernate/JPA, JUnit 5, Mockito, Mutiny (Reactive)

Build & Version Control: Maven, Gradle, Git, GitLab

Databases: PostgreSQL, SQL Server, MySQL, MongoDB, Redis

Cloud & DevOps: Docker, Kubernetes, GitLab CI/CD, Jenkins, GitHub Actions, AWS, Azure

Monitoring & Observability: Grafana, Prometheus, Kibana (ELK Stack)

API Design: RESTful APIs, OpenAPI/Swagger, API versioning and contract design